Melting Point of chemical molecules using Graph Neural Network

Made use of BradleyMeltingPointDataset for the model.

Downloading necessary libraries

```
!pip install "numpy<2.0"
```

```
Requirement already satisfied: numpy<2.0 in /usr/local/lib/python3.11/dist-packages (1.26.4)
```

```
!pip install rdkit-pypi
```

```
Collecting rdkit-pypi
  Using cached rdkit_pypi-2022.9.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.9 kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from rdkit-pypi) (1.26.4)
Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages (from rdkit-pypi) (11.3.0)
Downloading rdkit_pypi-2022.9.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (29.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 29.4/29.4 MB 51.8 MB/s eta 0:00:00
Installing collected packages: rdkit-pypi
Successfully installed rdkit-pypi-2022.9.5
```

```
!pip install torch torchvision torchaudio
```

```
Requirement already satisfied: torch in /usr/local/lib/python3.11/dist-packages (2.6.0+cu124)
Requirement already satisfied: torchvision in /usr/local/lib/python3.11/dist-packages (0.21.0+cu124)
Requirement already satisfied: torchaudio in /usr/local/lib/python3.11/dist-packages (2.6.0+cu124)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from torch) (3.18.0)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.11/dist-packages (from torch) (
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch) (3.5)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch) (3.1.6)
```

```
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from torch) (2025.3.0)
Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch)
  Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch)
  Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cuda-cupti-cu12==12.4.127 (from torch)
  Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch)
  Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cublas-cu12==12.4.5.8 (from torch)
  Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cufft-cu12==11.2.1.3 (from torch)
  Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-curand-cu12==10.3.5.147 (from torch)
  Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch)
  Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Collecting nvidia-cusparse-cu12==12.3.1.170 (from torch)
  Downloading nvidia_cusparse_cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages (from torc
Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch) (2
Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch)
Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch)
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Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages (from torch) (3.2.0)
Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages (from torch) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from torchvision) (1.26.4)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.11/dist-packages (from torchvision)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch) (3.
Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl (363.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 363.4/363.4 MB 4.2 MB/s eta 0:00:00
Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (13.8 MB)
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Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (24.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 24.6/24.6 MB 32.5 MB/s eta 0:00:00
Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (883 kB)
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Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl (664.8 MB)
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```

```
Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl (211.5 MB)
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Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl (56.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 56.3/56.3 MB 15.6 MB/s eta 0:00:00
Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl (127.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 127.9/127.9 MB 7.6 MB/s eta 0:00:00
Downloading nvidia_cusparse_cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl (207.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 207.5/207.5 MB 6.2 MB/s eta 0:00:00
Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (21.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 21.1/21.1 MB 69.9 MB/s eta 0:00:00
Installing collected packages: nvidia-nvjitlink-cu12, nvidia-curand-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-c
```

```
# Install PyTorch Geometric dependencies and core
!pip install torch-scatter -f https://data.pyg.org/whl/torch-2.2.0+cpu.html
!pip install torch-sparse -f https://data.pyg.org/whl/torch-2.2.0+cpu.html
!pip install torch-cluster -f https://data.pyg.org/whl/torch-2.2.0+cpu.html
!pip install torch-spline-conv -f https://data.pyg.org/whl/torch-2.2.0+cpu.html
!pip install torch-geometric
```

```
Looking in links: https://data.pyg.org/whl/torch-2.2.0+cpu.html
Collecting torch-scatter
  Downloading https://data.pyg.org/whl/torch-2.2.0%2Bcpu/torch_scatter-2.1.2%2Bpt22cpu-cp311-cp311-linux_x86_64.wh
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 511.8/511.8 kB 21.4 MB/s eta 0:00:00
Installing collected packages: torch-scatter
Successfully installed torch-scatter-2.1.2+pt22cpu
Looking in links: https://data.pyg.org/whl/torch-2.2.0+cpu.html
Collecting torch-sparse
  Downloading https://data.pyg.org/whl/torch-2.2.0%2Bcpu/torch_sparse-0.6.18%2Bpt22cpu-cp311-cp311-linux_x86_64.wh
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 45.4 MB/s eta 0:00:00
Requirement already satisfied: scipy in /usr/local/lib/python3.11/dist-packages (from torch-sparse) (1.16.0)
Requirement already satisfied: numpy<2.6,>=1.25.2 in /usr/local/lib/python3.11/dist-packages (from scipy->torch-sp
Installing collected packages: torch-sparse
Successfully installed torch-sparse-0.6.18+pt22cpu
Looking in links: https://data.pyg.org/whl/torch-2.2.0+cpu.html
Collecting torch-cluster
  Downloading https://data.pyg.org/whl/torch-2.2.0%2Bcpu/torch_cluster-1.6.3%2Bpt22cpu-cp311-cp311-linux_x86_64.wh
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 776.6/776.6 kB 23.1 MB/s eta 0:00:00
Requirement already satisfied: scipy in /usr/local/lib/python3.11/dist-packages (from torch-cluster) (1.16.0)
Requirement already satisfied: numpy<2.6,>=1.25.2 in /usr/local/lib/python3.11/dist-packages (from scipy->torch-cl
```

```
Installing collected packages: torch-cluster
Successfully installed torch-cluster-1.6.3+pt22cpu
Looking in links: https://data.pyg.org/whl/torch-2.2.0+cpu.html
Collecting torch-spline-conv
  Downloading https://data.pyg.org/whl/torch-2.2.0%2Bcpu/torch_spline_conv-1.2.2%2Bpt22cpu-cp311-cp311-linux_x86_6
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 215.8/215.8 kB 12.4 MB/s eta 0:00:00
Installing collected packages: torch-spline-conv
Successfully installed torch-spline-conv-1.2.2+pt22cpu
Collecting torch-geometric
  Downloading torch_geometric-2.6.1-py3-none-any.whl.metadata (63 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 63.1/63.1 kB 4.7 MB/s eta 0:00:00
Requirement already satisfied: aiohttp in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (3.12.14)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (2025.3.0)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (3.1.6)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (1.26.4)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (5.
Requirement already satisfied: pyparsing in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (3.2.3)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (2.32.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from torch-geometric) (4.67.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->t
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->torch-ge
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->torch-geome
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp->torch-g
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp->torch
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->torch-ge
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->torch-g
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch-geom
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests-
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests->torch-geome
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests->torch
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests->torch
Requirement already satisfied: typing-extensions>=4.2 in /usr/local/lib/python3.11/dist-packages (from aiosignal>=
Downloading torch_geometric-2.6.1-py3-none-any.whl (1.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.1/1.1 MB 60.8 MB/s eta 0:00:00
Installing collected packages: torch-geometric
Successfully installed torch-geometric-2.6.1
```

```
!nvidia-smi
```

```
Wed Jul 30 16:27:46 2025
```

```
Wed Jul 30 16:27:46 2025
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 550.54.15              Driver Version: 550.54.15      CUDA Version: 12.4     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   47C    P8             10W /   70W |       0MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                              GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

```
## Importing necessary libraries
import pandas as pd
import rdkit
import scipy
from rdkit import Chem
from rdkit.Chem import Draw
import torch
import matplotlib.pyplot as plt
import torch_geometric
```

```
/usr/local/lib/python3.11/dist-packages/torch_geometric/typing.py:86: UserWarning: An issue occurred while importi
  warnings.warn(f"An issue occurred while importing 'torch-scatter'. "
/usr/local/lib/python3.11/dist-packages/torch_geometric/typing.py:97: UserWarning: An issue occurred while importi
  warnings.warn(f"An issue occurred while importing 'torch-cluster'. "
/usr/local/lib/python3.11/dist-packages/torch_geometric/typing.py:113: UserWarning: An issue occurred while import
  warnings.warn(
/usr/local/lib/python3.11/dist-packages/torch_geometric/typing.py:124: UserWarning: An issue occurred while import
  warnings.warn(f"An issue occurred while importing 'torch-sparse'. "
```

Uploading the dataset from google drive

```
from google.colab import drive
drive.mount('/content/gdrive')
```

```
Mounted at /content/gdrive
```

```
df = pd.read_csv("/content/gdrive/MyDrive/MeltingPointDataset.csv")
df = df[['name','smiles','mpC']]
df
```

	name	smiles	mpC
0	2-(2,4-dinitrobenzyl)pyridine	c1ccnc(c1)Cc2ccc(cc2[N+](=O)[O-])[N+](=O)[O-]	92.0
1	2-(1-piperidinyl)aniline	c1ccc(c(c1)N)N2CCCCC2	46.0
2	2-(1-piperazinyl)pyrimidine	c1cnc(nc1)N2CCNCC2	33.0
3	2-(1-piperazinyl)phenol	c1ccc(c(c1)N2CCNCC2)O	125.0
4	2-(1-cyclohexenyl)ethylamine	C1CCC(=CC1)CCN	-55.0
...	...	...	...
28640	xanthene	c1ccc2c(c1)Cc3ccccc3O2	100.5
28641	xanthene	c1ccc2c(c1)Cc3ccccc3O2	100.5
28642	xanthene	c1ccc2c(c1)Cc3ccccc3O2	101.5
28643	Xanthene	C1C2=CC=CC=C2OC3=CC=CC=C31	101.0
28644	Z-crotononitrile	N#C\C=C/C	-72.6

28645 rows × 3 columns

28645 rows × 3 columns

```
df
```

	name	smiles	mpC
0	2-(2,4-dinitrobenzyl)pyridine	c1ccnc(c1)Cc2ccc(cc2[N+](=O)[O-])[N+](=O)[O-]	92.0
1	2-(1-piperidinyl)aniline	c1ccc(c(c1)N)N2CCCCC2	46.0
2	2-(1-piperazinyl)pyrimidine	c1cnc(nc1)N2CCNCC2	33.0
3	2-(1-piperazinyl)phenol	c1ccc(c(c1)N2CCNCC2)O	125.0
4	2-(1-cyclohexenyl)ethylamine	C1CCC(=CC1)CCN	-55.0
...	...	...	...
28640	xanthene	c1ccc2c(c1)Cc3ccccc3O2	100.5
28641	xanthene	c1ccc2c(c1)Cc3ccccc3O2	100.5
28642	xanthene	c1ccc2c(c1)Cc3ccccc3O2	101.5
28643	Xanthene	C1C2=CC=CC=C2OC3=CC=CC=C31	101.0
28644	Z-crotononitrile	N#C\C=C/C	-72.6

28645 rows × 3 columns

```
df['name'] = df['name'].str.lower()
df['name'].duplicated().sum()/len(df)
```

```
0.16201780415430267
```

```
df['smiles'].duplicated().sum()/len(df)
```

```
0.16222726479315763
```

```
df_new= df.drop_duplicates(subset=['name','smiles'], keep='first', inplace=False, ignore_index=False)
```

```
df_new.shape
```

```
(26244, 3)
```

```
df_new
```

	name	smiles	mpC
0	2-(2,4-dinitrobenzyl)pyridine	c1ccnc(c1)Cc2ccc(cc2[N+](=O)[O-])[N+](=O)[O-]	92.0
1	2-(1-piperidinyl)aniline	c1ccc(c(c1)N)N2CCCCC2	46.0
2	2-(1-piperazinyl)pyrimidine	c1cnc(nc1)N2CCNCC2	33.0
3	2-(1-piperazinyl)phenol	c1ccc(c(c1)N2CCNCC2)O	125.0
4	2-(1-cyclohexenyl)ethylamine	C1CCC(=CC1)CCN	-55.0
...	...	...	...
28634	veratraldehyde	COc1cc(ccc1OC)C=O	43.0
28635	warfarin	CC(=O)CC(C\1=C(/O)c2ccccc2OC/1=O)c3ccccc3	161.0
28640	xanthene	c1ccc2c(c1)Cc3ccccc3O2	100.5
28643	xanthene	C1C2=CC=CC=C2OC3=CC=CC=C31	101.0
28644	z-crotononitrile	N#C\C=C/C	-72.6

26244 rows × 3 columns

Cleaning up the datasets

We are kekulising the smiles so as to use it

```
def kekule_smiles(smiles):
    try:
        mol = Chem.MolFromSmiles(smiles, sanitize=False)
        Chem.SanitizeMol(mol)  # Run checks
        Chem.Kekulize(mol, clearAromaticFlags=True)
        return Chem.MolToSmiles(mol, kekuleSmiles=True)
    except Exception as e:
        return None  # or log: f"{smiles} --> {e}"
```

```
df_new["kekule"] = df_new["smiles"].apply(kekule_smiles)
df_new['kekule']
```

```
[16:29:06] Can't kekulize mol.  Unkekulized atoms: 6 7 8 9 11 12 13 15 16 17 18 19 22 23 24
[16:29:06] Can't kekulize mol.  Unkekulized atoms: 1 2 3 4 5 7 8 10 11 12 13 15 16 17 19
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[16:29:06] Explicit valence for atom # 20 C, 5, is greater than permitted
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[16:29:06] Can't kekulize mol.  Unkekulized atoms: 0 1 2 3 4 5 6 7 8
[16:29:06] Can't kekulize mol.  Unkekulized atoms: 0 1 2 3 4
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[16:29:06] Can't kekulize mol.  Unkekulized atoms: 1 2 3 4 5 6 8 9 10 11 12 13 14
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[16:29:06] Can't kekulize mol.  Unkekulized atoms: 0 1 2 3 12 13 14 15 16
[16:29:06] Explicit valence for atom # 1 Br, 5, is greater than permitted
[16:29:06] Explicit valence for atom # 1 Br, 3, is greater than permitted
[16:29:06] Explicit valence for atom # 1 Cl, 3, is greater than permitted
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[16:29:08] Can't kekulize mol.  Unkekulized atoms: 1 2 3 4 5
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 16 17 18 19 20 21 22 23 24
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 2 3 4 5 6 7 20 21 22
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 2 3 4 5 6 7 8 9 10 11 12 13 14
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 1 2 3 4 5 6 7 8 9 10 11 12 13
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 11 12 13 14 15 16 17 20 21
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 14 15 16 17 18 19 20 21 22
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 13 14 15 16 17 18 19 20 21
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 2 3 4 5 6 7 8 9 10
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 6 7 8 9 10 11 12 13 18
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 1 2 3 4 5 6 8 9 10
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 9 10 11 12 13 14 15 16 17
[16:29:08] Explicit valence for atom # 6 N, 5, is greater than permitted
[16:29:08] Explicit valence for atom # 20 N, 5, is greater than permitted
```

```
[16:29:08] Explicit valence for atom # 8 N, 4, is greater than permitted
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 7 8 9 10 11 12 13 14 15
[16:29:08] Explicit valence for atom # 1 Cl, 2, is greater than permitted
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 3 4 5 6 7
[16:29:08] Explicit valence for atom # 1 N, 4, is greater than permitted
[16:29:08] Explicit valence for atom # 4 N, 4, is greater than permitted
[16:29:08] Explicit valence for atom # 1 N, 4, is greater than permitted
[16:29:08] Explicit valence for atom # 5 N, 4, is greater than permitted
[16:29:08] Explicit valence for atom # 1 B, 4, is greater than permitted
[16:29:08] Explicit valence for atom # 12 N, 4, is greater than permitted
[16:29:08] Explicit valence for atom # 10 N, 4, is greater than permitted
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 3 4 5 6 7 8 9 10 11
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 4
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 4 5 8 9 10
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 0 1 2 3 4
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 3 4 5 6 7
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 0 1 2 3 4 5 6 7 8
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 5 6 7 8 9 10 11 12 13
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 5 6 7 8 9
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 4 5 6 7 8 9 10 11 12
[16:29:08] Can't kekulize mol.  Unkekulized atoms: 6 7 8 9 10
/tmp/ipython-input-16-2126998567.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returr
  df_new["kekule"] = df_new["smiles"].apply(kekule_smiles)
```

	kekule
0	O=[N+]([O-])C1=CC=C(CC2=CC=CC=N2)C([N+](=O)[O-...
1	NC1=CC=CC=C1N1CCCCC1
2	C1=CN=C(N2CCNCC2)N=C1
3	OC1=CC=CC=C1N1CCNCC1
4	NCCC1=CCCCC1
...	...

28634	COC1=CC=C(C=O)C=C1OC
28635	CC(=O)CC(C1=CC=CC=C1)C1=C(O)C2=CC=CC=C2OC1=O
28640	C1=CC=C2OC3=CC=CC=C3CC2=C1
28643	C1=CC=C2OC3=CC=CC=C3CC2=C1
28644	C/C=C\C#N

26244 rows × 1 columns

dtype: object

```
null_count = df_new['kekule'].isna().sum()
print(f"Number of null values in the kekule column: {null_count}")
```

```
Number of null values in the kekule column: 279
```

Dropping the null values from the kekulised smiles column

```
df_new = df_new.dropna(subset=['kekule'])
df_new
```

	name	smiles	mpC	kekule
0	2-(2,4-dinitrobenzyl)pyridine	c1ccnc(c1)Cc2ccc(cc2[N+](=O)[O-])[N+](=O)[O-]	92.0	O=[N+]([O-])C1=CC=C(CC2=CC=CC=N2)C([N+](=O)[O-...
1	2-(1-piperidinyl)aniline	c1ccc(c(c1)N)N2CCCCC2	46.0	NC1=CC=CC=C1N1CCCCC1
2	2-(1-piperazinyl)pyrimidine	c1cnc(nc1)N2CCNCC2	33.0	C1=CN=C(N2CCNCC2)N=C1
3	2-(1-piperazinyl)phenol	c1ccc(c(c1)N2CCNCC2)O	125.0	OC1=CC=CC=C1N1CCNCC1

4	2-(1-cyclohexenyl)ethylamine	C1CCC(=CC1)CCN	-55.0	NCCC1=CCCCC1
...	...	...	...	...
28634	veratraldehyde	COc1cc(ccc1OC)C=O	43.0	COC1=CC=C(C=O)C=C1OC
28635	warfarin	CC(=O)CC(C\1=C(/O)c2ccccc2OC/1=O)c3ccccc3	161.0	CC(=O)CC(C1=CC=CC=C1)C1=C(O)C2=CC=CC=C2OC1=O
28640	xanthene	c1ccc2c(c1)Cc3ccccc3O2	100.5	C1=CC=C2OC3=CC=CC=C3CC2=C1
28643	xanthene	C1C2=CC=CC=C2OC3=CC=CC=C31	101.0	C1=CC=C2OC3=CC=CC=C3CC2=C1
28644	z-crotononitrile	N#C\C=C/C	-72.6	C/C=C\C#N

```
df_new["kekule"][0]
```

```
'O=[N+]([O-])C1=CC=C(CC2=CC=CC=N2)C([N+](=O)[O-])=C1'
```

converting the smiles to mol file

```
df_new["Mol"] = df_new["kekule"].apply(Chem.MolFromSmiles)
df_new['Mol']
```

```
/tmp/ipython-input-20-2178509535.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returr
  df_new["Mol"] = df_new["kekule"].apply(Chem.MolFromSmiles)
```

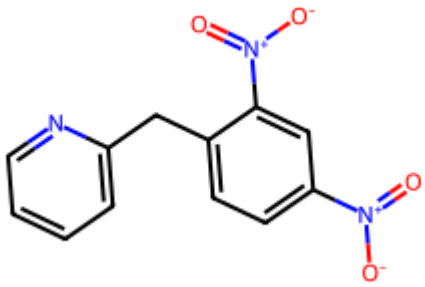
	Mol
0	<rdkit.Chem.rdchem.Mol object at 0x7fec003a2b20>
1	<rdkit.Chem.rdchem.Mol object at 0x7fec003a2490>
2	<rdkit.Chem.rdchem.Mol object at 0x7fec003a3060>

3	<rdkit.Chem.rdchem.Mol object at 0x7fec003a2c00>
4	<rdkit.Chem.rdchem.Mol object at 0x7fec003a26c0>
...	...
28634	<rdkit.Chem.rdchem.Mol object at 0x7fec00092880>
28635	<rdkit.Chem.rdchem.Mol object at 0x7fec000928f0>
28640	<rdkit.Chem.rdchem.Mol object at 0x7fec00092960>
28643	<rdkit.Chem.rdchem.Mol object at 0x7fec000929d0>
28644	<rdkit.Chem.rdchem.Mol object at 0x7fec00092a40>

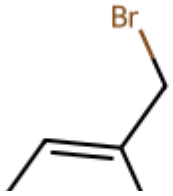
25965 rows × 1 columns

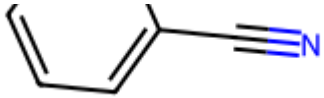
dtype: object

```
df_new["Mol"][0]
```

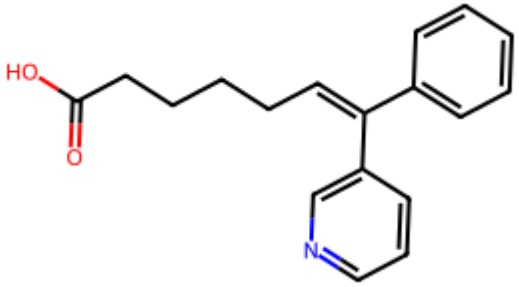



```
df_new["Mol"][1000]
```

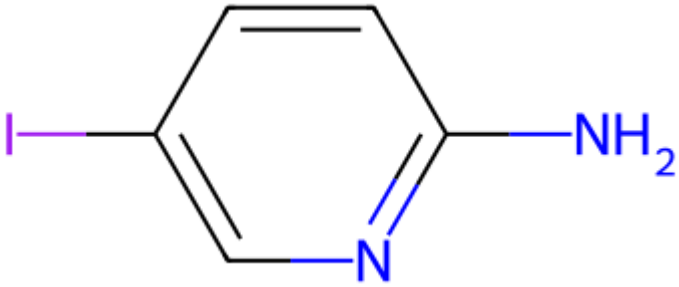





```
df_new["Mol"][10000]
```




```
df_new["Mol"][2000]
```



Indexing the diffrent atoms present in a molecule

```
def get_idx(molecule):
    for atom in molecule.GetAtoms():
        #print (atom)
        idx_atom = atom.GetIdx()
        #print(idx_atom)
        atom.SetAtomMapNum(idx_atom)
```

```
    return (molecule)

df_new["idx_mol"] = df_new["Mol"].apply(lambda x: get_idx(x))
df_new["idx_mol"][15892]
```

```
/tmp/ipython-input-25-2999795306.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returr
  df_new["idx_mol"] = df_new["Mol"].apply(lambda x: get_idx(x))
```

Finding out the edges of the molecule

```
def find_edge(molecule):
    edges = []
    for bond in molecule.GetBonds():
        i = bond.GetBeginAtomIdx()
        j = bond.GetEndAtomIdx()
        edges.append([i,j])
        edges.append([j,i])
    return edges

df_new['edge'] = df_new['idx_mol'].apply(lambda x:find_edge(x))
df_new["edge"][15892]
```

```
/tmp/ipython-input-26-3664346821.py:10: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
```

```
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returr
  df_new['edge'] = df_new['idx_mol'].apply(lambda x:find_edge(x))
[]
```

Gettign the adjacency matrix

```
from rdkit.Chem import rdmolops
```

```
def adj_matrix(molecule):
    x = rdkit.Chem.rdmolops.GetAdjacencyMatrix(molecule)

    return torch.tensor(x, dtype = torch.long)# check with google tensor?

df_new["adj_matrix"] = df_new["idx_mol"].apply(lambda x: adj_matrix(x))
df_new["adj_matrix"][2000]
```

```
/tmp/ipython-input-28-646405076.py:6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returr
  df_new["adj_matrix"] = df_new["idx_mol"].apply(lambda x: adj_matrix(x))
tensor([[0, 1, 0, 0, 0, 0, 0, 0],
        [1, 0, 1, 0, 0, 0, 0, 1],
        [0, 1, 0, 1, 0, 0, 0, 0],
        [0, 0, 1, 0, 1, 0, 0, 0],
        [0, 0, 0, 1, 0, 1, 1, 0],
        [0, 0, 0, 0, 1, 0, 0, 0],
        [0, 0, 0, 0, 1, 0, 0, 1],
        [0, 1, 0, 0, 0, 0, 1, 0]])
```

```
df_new["edge"][1]
```

```
[[0, 1],
 [1, 0],
 [1, 2],
 [2, 1],
 [2, 3],
 [3, 2],
 [3, 4],
 [4, 3],
 [4, 5],
 [5, 4],
 [5, 6],
 [6, 5],
 [6, 7],
 [7, 6],
 [7, 8],
 [8, 7],
 [8, 9],
 [9, 8],
 [9, 10],
 [10, 9],
 [10, 11],
 [11, 10],
 [11, 12],
 [12, 11],
 [6, 1],
 [1, 6],
 [12, 7],
 [7, 12]]
```

```
# x = list(zip([1,2],[2,1],[2,3],[3,4]))
# x
```

```
# row, col = list(zip(*df_new["edge"][2000]))
# #print(row)
# #print(col)
# x = torch.tensor([row, col])
# print(x)
```

```
empty_edges_mask = df_new["edge"].apply(lambda x: len(x) == 0)
empty_edges_mask
empty_rows = df_new[empty_edges_mask]
#empty_rows.head()

print(f"Number of graphs with empty edges: {len(empty_rows)}")
print(empty_rows)
```

```
Number of graphs with empty edges: 12
                     name       smiles     mpC       kekule  \
15655            caesium         [Cs]   28.50         [Cs]
15852   hydrogen bromide           Br  -87.00           Br
15854  hydrogen chloride           Cl -114.20           Cl
15892            methane            C -182.00            C
17699         phosphorus            P   44.10            P
17700          phosphine            P -133.00            P
18555  hydrogen fluoride            F  -83.55            F
22330              water            O    0.00            O
22450   hydrogen sulfide            S  -85.49            S
22490            ammonia            N  -77.70            N
25711             sulfur            S  112.00            S
28528    sodium chloride  [Na+].[Cl-]  804.00  [Cl-].[Na+]

                                                     Mol  \
15655  <rdkit.Chem.rdchem.Mol object at 0x7fec00169b60>
15852  <rdkit.Chem.rdchem.Mol object at 0x7fec0016dfc0>
15854  <rdkit.Chem.rdchem.Mol object at 0x7fec0016e030>
15892  <rdkit.Chem.rdchem.Mol object at 0x7fec0016eea0>
17699  <rdkit.Chem.rdchem.Mol object at 0x7fec00197a70>
17700  <rdkit.Chem.rdchem.Mol object at 0x7fec00197ae0>
18555  <rdkit.Chem.rdchem.Mol object at 0x7fec001acdd0>
22330  <rdkit.Chem.rdchem.Mol object at 0x7fec001fcac0>
22450  <rdkit.Chem.rdchem.Mol object at 0x7fec001ff060>
22490  <rdkit.Chem.rdchem.Mol object at 0x7fec001ffc30>
25711  <rdkit.Chem.rdchem.Mol object at 0x7fec00058970>
28528  <rdkit.Chem.rdchem.Mol object at 0x7fec00090a50>
```

```
                                                idx_mol edge  \
15655  <rdkit.Chem.rdchem.Mol object at 0x7fec00169b60>   []
15852  <rdkit.Chem.rdchem.Mol object at 0x7fec0016dfc0>   []
15854  <rdkit.Chem.rdchem.Mol object at 0x7fec0016e030>   []
15892  <rdkit.Chem.rdchem.Mol object at 0x7fec0016eea0>   []
17699  <rdkit.Chem.rdchem.Mol object at 0x7fec00197a70>   []
17700  <rdkit.Chem.rdchem.Mol object at 0x7fec00197ae0>   []
18555  <rdkit.Chem.rdchem.Mol object at 0x7fec001acdd0>   []
22330  <rdkit.Chem.rdchem.Mol object at 0x7fec001fcac0>   []
22450  <rdkit.Chem.rdchem.Mol object at 0x7fec001ff060>   []
22490  <rdkit.Chem.rdchem.Mol object at 0x7fec001ffc30>   []
25711  <rdkit.Chem.rdchem.Mol object at 0x7fec00058970>   []
28528  <rdkit.Chem.rdchem.Mol object at 0x7fec00090a50>   []

                                                adj_matrix
15655                                       [[tensor(0)]]
15852                                       [[tensor(0)]]
15854                                       [[tensor(0)]]
15892                                       [[tensor(0)]]
17699                                       [[tensor(0)]]
17700                                       [[tensor(0)]]
18555                                       [[tensor(0)]]
22330                                       [[tensor(0)]]
22450                                       [[tensor(0)]]
22490                                       [[tensor(0)]]
25711                                       [[tensor(0)]]
28528  [[tensor(0), tensor(0)], [tensor(0), tensor(0)]]
```

Converting the edge index to COOformat

```
def coo_edge_index(edges):
    if edges:
        row, col = list(zip(*edges))
        indices = torch.tensor([row, col], dtype=torch.long)
        return indices
    else:
        return torch.empty((2, 0), dtype=torch.long)  # Return empty edge index if no edges
```

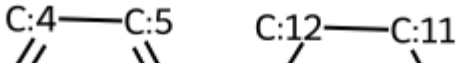
```
df_new["coo_matrix"] = df_new["edge"].apply(coo_edge_index)
```

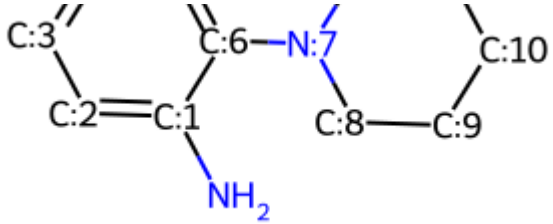
```
/tmp/ipython-input-33-185872523.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returr
  df_new["coo_matrix"] = df_new["edge"].apply(coo_edge_index)
```

```
df_new.head()
```

	name	smiles	mpC	kekule	Mol	
0	2-(2,4-dinitrobenzyl)pyridine	c1ccnc(c1)Cc2ccc(cc2[N+](=O)[O-])[N+](=O)[O-]	92.0	O=[N+]([O-])C1=CC=C(CC2=CC=CC=N2)C([N+](=O)[O-...	<rdkit.Chem.rdchem.Mol object at 0x7fec003a2b20>	<rdkit.Chem.r 0x7fec
1	2-(1-piperidinyl)aniline	c1ccc(c(c1)N)N2CCCCC2	46.0	NC1=CC=CC=C1N1CCCCC1	<rdkit.Chem.rdchem.Mol object at	<rdkit.Chem.r

```
df_new["Mol"][1]
```





Getting the node feature matrix: feature 1: atomic number; feature 2: degree of the atoms; feature 3: Number of implicit hydrogen; feature 4: checking whether the atom is part of ring and feature 5: whether atom is aromatic or not

```
def node_feature(atom):
    return [atom.GetAtomicNum(),
          atom.GetDegree(),
          atom.GetNumImplicitHs(),
            atom.IsInRing(),
            atom.GetIsAromatic()]

def node_feature_matrix(molecule):
    node_matrix = []
    for atoms in molecule.GetAtoms():
        node_matrix.append(node_feature(atoms))
    return torch.tensor(node_matrix, dtype = torch.float)

df_new["node_matrix"] = df_new["idx_mol"].apply(lambda x: node_feature_matrix(x))
#df.head()
df_new["node_matrix"][1]
```

```
/tmp/ipython-input-36-3931165104.py:14: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returr
  df_new["node_matrix"] = df_new["idx_mol"].apply(lambda x: node_feature_matrix(x))
tensor([[7., 1., 2., 0., 0.],
        [6., 3., 0., 1., 1.],
        [6., 2., 1., 1., 1.],
```

```
       [6., 2., 1., 1., 1.],
       [6., 2., 1., 1., 1.],
       [6., 2., 1., 1., 1.],
       [6., 2., 1., 1., 1.],
       [6., 3., 0., 1., 1.],
       [7., 3., 0., 1., 0.],
       [6., 2., 2., 1., 0.],
       [6., 2., 2., 1., 0.],
       [6., 2., 2., 1., 0.],
       [6., 2., 2., 1., 0.],
       [6., 2., 2., 1., 0.]])
```

```
# first_col = df_new["node_matrix"][1][:,0]
# first_col
```

```
tensor([7., 6., 6., 6., 6., 6., 6., 7., 6., 6., 6., 6., 6.])
```

```
df_new["node_matrix"]
```

	node_matrix
0	[[tensor(8.), tensor(1.), tensor(0.), tensor(0...
1	[[tensor(7.), tensor(1.), tensor(2.), tensor(0...
2	[[tensor(6.), tensor(2.), tensor(1.), tensor(1...
3	[[tensor(8.), tensor(1.), tensor(1.), tensor(0...
4	[[tensor(7.), tensor(1.), tensor(2.), tensor(0...
...	...
28634	[[tensor(6.), tensor(1.), tensor(3.), tensor(0...
28635	[[tensor(6.), tensor(1.), tensor(3.), tensor(0...
28640	[[tensor(6.), tensor(2.), tensor(1.), tensor(1...
28643	[[tensor(6.), tensor(2.), tensor(1.), tensor(1...

28644 [[tensor(6.), tensor(1.), tensor(3.), tensor(0...

25965 rows × 1 columns

dtype: object

```
#for matrix in df_new["node_matrix"]:
max_val = max(matrix[:, 0].max().item() for matrix in df_new['node_matrix'])
print("The highest atomic number present in the dataset is:", max_val)
```

```
The highest atomic number present in the dataset is: 82.0
```

```
for i in range(1,83):
  count = sum((t[:, 0] == i).any().item() for t in df_new["node_matrix"])

  print(f"Number of compounds with element of atomic number {i} present is: {count}")
```

```
Number of compounds with element of atomic number 1 present is: 21
Number of compounds with element of atomic number 2 present is: 0
Number of compounds with element of atomic number 3 present is: 0
Number of compounds with element of atomic number 4 present is: 0
Number of compounds with element of atomic number 5 present is: 241
Number of compounds with element of atomic number 6 present is: 25890
Number of compounds with element of atomic number 7 present is: 13370
Number of compounds with element of atomic number 8 present is: 19766
Number of compounds with element of atomic number 9 present is: 2473
Number of compounds with element of atomic number 10 present is: 0
Number of compounds with element of atomic number 11 present is: 2
Number of compounds with element of atomic number 12 present is: 1
Number of compounds with element of atomic number 13 present is: 1
Number of compounds with element of atomic number 14 present is: 199
Number of compounds with element of atomic number 15 present is: 420
Number of compounds with element of atomic number 16 present is: 3061
Number of compounds with element of atomic number 17 present is: 4216
```

```
Number of compounds with element of atomic number 18 present is: 0
Number of compounds with element of atomic number 19 present is: 0
Number of compounds with element of atomic number 20 present is: 0
Number of compounds with element of atomic number 21 present is: 0
Number of compounds with element of atomic number 22 present is: 1
Number of compounds with element of atomic number 23 present is: 0
Number of compounds with element of atomic number 24 present is: 1
Number of compounds with element of atomic number 25 present is: 0
Number of compounds with element of atomic number 26 present is: 0
Number of compounds with element of atomic number 27 present is: 1
Number of compounds with element of atomic number 28 present is: 0
Number of compounds with element of atomic number 29 present is: 0
Number of compounds with element of atomic number 30 present is: 1
Number of compounds with element of atomic number 31 present is: 0
Number of compounds with element of atomic number 32 present is: 1
Number of compounds with element of atomic number 33 present is: 4
Number of compounds with element of atomic number 34 present is: 7
Number of compounds with element of atomic number 35 present is: 1846
Number of compounds with element of atomic number 36 present is: 0
Number of compounds with element of atomic number 37 present is: 0
Number of compounds with element of atomic number 38 present is: 0
Number of compounds with element of atomic number 39 present is: 0
Number of compounds with element of atomic number 40 present is: 0
Number of compounds with element of atomic number 41 present is: 0
Number of compounds with element of atomic number 42 present is: 0
Number of compounds with element of atomic number 43 present is: 0
Number of compounds with element of atomic number 44 present is: 0
Number of compounds with element of atomic number 45 present is: 0
Number of compounds with element of atomic number 46 present is: 0
Number of compounds with element of atomic number 47 present is: 1
Number of compounds with element of atomic number 48 present is: 0
Number of compounds with element of atomic number 49 present is: 0
Number of compounds with element of atomic number 50 present is: 0
Number of compounds with element of atomic number 51 present is: 0
Number of compounds with element of atomic number 52 present is: 0
Number of compounds with element of atomic number 53 present is: 473
Number of compounds with element of atomic number 54 present is: 0
Number of compounds with element of atomic number 55 present is: 1
Number of compounds with element of atomic number 56 present is: 0
Number of compounds with element of atomic number 57 present is: 0
```

```
Number of compounds with element of atomic number 58 present is: 0
```

```
df_new["node_matrix"][0]
```

```
tensor([[8., 1., 0., 0., 0.],
        [7., 3., 0., 0., 0.],
        [8., 1., 0., 0., 0.],
        [6., 3., 0., 1., 1.],
        [6., 2., 1., 1., 1.],
        [6., 2., 1., 1., 1.],
        [6., 3., 0., 1., 1.],
        [6., 2., 2., 0., 0.],
        [6., 3., 0., 1., 1.],
        [6., 2., 1., 1., 1.],
        [6., 2., 1., 1., 1.],
        [6., 2., 1., 1., 1.],
        [6., 2., 1., 1., 1.],
        [7., 2., 0., 1., 1.],
        [6., 3., 0., 1., 1.],
        [7., 3., 0., 0., 0.],
        [8., 1., 0., 0., 0.],
        [8., 1., 0., 0., 0.],
        [6., 2., 1., 1., 1.]])
```

```
len(df_new)
```

```
25965
```

```
import torch_geometric
from torch_geometric.data import Data
```

Creating dataset for the model

```
dataset = []
for _, row in df_new.iterrows():
```

```
for _, row in df_new.iterrows():
    x = row["node_matrix"]
    edge_index = row["coo_matrix"]
    y = torch.tensor([row["mpC"]], dtype=torch.float)

    data = Data(x=x, edge_index=edge_index, y=y)
    dataset.append(data)

dataset
```

```
[Data(x=[19, 5], edge_index=[2, 40], y=[1]),
 Data(x=[13, 5], edge_index=[2, 28], y=[1]),
 Data(x=[12, 5], edge_index=[2, 26], y=[1]),
 Data(x=[13, 5], edge_index=[2, 28], y=[1]),
 Data(x=[9, 5], edge_index=[2, 18], y=[1]),
 Data(x=[19, 5], edge_index=[2, 38], y=[1]),
 Data(x=[21, 5], edge_index=[2, 44], y=[1]),
 Data(x=[20, 5], edge_index=[2, 40], y=[1]),
 Data(x=[7, 5], edge_index=[2, 12], y=[1]),
 Data(x=[11, 5], edge_index=[2, 24], y=[1]),
 Data(x=[18, 5], edge_index=[2, 42], y=[1]),
 Data(x=[4, 5], edge_index=[2, 6], y=[1]),
 Data(x=[12, 5], edge_index=[2, 26], y=[1]),
 Data(x=[13, 5], edge_index=[2, 28], y=[1]),
 Data(x=[12, 5], edge_index=[2, 26], y=[1]),
 Data(x=[12, 5], edge_index=[2, 26], y=[1]),
 Data(x=[10, 5], edge_index=[2, 20], y=[1]),
 Data(x=[5, 5], edge_index=[2, 8], y=[1]),
 Data(x=[16, 5], edge_index=[2, 30], y=[1]),
 Data(x=[9, 5], edge_index=[2, 16], y=[1]),
 Data(x=[8, 5], edge_index=[2, 14], y=[1]),
 Data(x=[9, 5], edge_index=[2, 16], y=[1]),
 Data(x=[12, 5], edge_index=[2, 22], y=[1]),
 Data(x=[10, 5], edge_index=[2, 18], y=[1]),
 Data(x=[6, 5], edge_index=[2, 10], y=[1]),
 Data(x=[12, 5], edge_index=[2, 24], y=[1]),
 Data(x=[14, 5], edge_index=[2, 28], y=[1]),
 Data(x=[21, 5], edge_index=[2, 44], y=[1]),
 Data(x=[22, 5], edge_index=[2, 46], y=[1]),
```

```
Data(x=[14, 5], edge_index=[2, 30], y=[1]),
Data(x=[13, 5], edge_index=[2, 28], y=[1]),
Data(x=[11, 5], edge_index=[2, 24], y=[1]),
Data(x=[14, 5], edge_index=[2, 30], y=[1]),
Data(x=[12, 5], edge_index=[2, 26], y=[1]),
Data(x=[13, 5], edge_index=[2, 26], y=[1]),
Data(x=[9, 5], edge_index=[2, 18], y=[1]),
Data(x=[7, 5], edge_index=[2, 14], y=[1]),
Data(x=[12, 5], edge_index=[2, 26], y=[1]),
Data(x=[11, 5], edge_index=[2, 24], y=[1]),
Data(x=[14, 5], edge_index=[2, 28], y=[1]),
Data(x=[15, 5], edge_index=[2, 30], y=[1]),
Data(x=[16, 5], edge_index=[2, 32], y=[1]),
Data(x=[14, 5], edge_index=[2, 30], y=[1]),
Data(x=[12, 5], edge_index=[2, 26], y=[1]),
Data(x=[13, 5], edge_index=[2, 28], y=[1]),
Data(x=[13, 5], edge_index=[2, 28], y=[1]),
Data(x=[14, 5], edge_index=[2, 30], y=[1]),
Data(x=[13, 5], edge_index=[2, 28], y=[1]),
Data(x=[13, 5], edge_index=[2, 28], y=[1]),
Data(x=[15, 5], edge_index=[2, 34], y=[1]),
Data(x=[11, 5], edge_index=[2, 20], y=[1]),
Data(x=[11, 5], edge_index=[2, 20], y=[1]),
Data(x=[12, 5], edge_index=[2, 22], y=[1]),
Data(x=[24, 5], edge_index=[2, 54], y=[1]),
Data(x=[10, 5], edge_index=[2, 20], y=[1]),
Data(x=[13, 5], edge_index=[2, 24], y=[1]),
Data(x=[24, 5], edge_index=[2, 48], y=[1]),
Data(x=[14, 5], edge_index=[2, 26], y=[1]),
```

Splitting into train and test dataset

```
import random

random.seed(13)
data = random.sample(dataset, len(dataset))
```

```
train_dataset = dataset[:25000]
test_dataset = dataset[25000:]

print(f'Number of training graphs: {len(train_dataset)}')
print(f'Number of test graphs: {len(test_dataset)}')
```

```
Number of training graphs: 25000
Number of test graphs: 965
```

Splitting to batches using dataloader

```
from torch_geometric.loader import DataLoader
BATCH_SIZE = 32
train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE, shuffle=False)

print(f"Dataloaders: {train_loader, test_loader}")
print(f"Length of train loader: {len(train_loader)} batches of {BATCH_SIZE}")
print(f"Length of test loader: {len(test_loader)} batches of {BATCH_SIZE}")


# for step, data in enumerate(train_loader):
#     print(f'Step {step + 1}:')
#     print('=======')
#     print(f'Number of graphs in the current training batch: {data.num_graphs}')
#     print(data)
#     print()

# for step, data in enumerate(test_loader):
#     print(f'Step {step + 1}:')
#     print('=======')
#     print(f'Number of graphs in the current testing batch: {data.num_graphs}')
#     print(data)
#     print()
```

```
Dataloaders: (<torch_geometric.loader.dataloader.DataLoader object at 0x7febf71888d0>, <torch_geometric.loader.dat
Length of train loader: 782 batches of 32
Length of test loader: 31 batches of 32
```

```
print(len(train_loader))
```

```
782
```

```
for i, batch in enumerate(train_loader):
    print(f"\n🔹 Batch {i+1}")
    print(f"Batch {batch}")
    print("Number of nodes:", batch.x.size(0))
    print("Number of edges:", batch.edge_index.size(1))
    print("Target shape:", batch.y.shape)
    print("Graphs in batch:", batch.num_graphs)
    print("Batch vector shape:", batch.batch.shape)

    if i == 4:   # Stop after 5 batches
        break
```

```
🔹 Batch 1
Batch DataBatch(x=[426, 5], edge_index=[2, 870], y=[32], batch=[426], ptr=[33])
Number of nodes: 426
Number of edges: 870
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([426])

🔹 Batch 2
Batch DataBatch(x=[482, 5], edge_index=[2, 1010], y=[32], batch=[482], ptr=[33])
Number of nodes: 482
Number of edges: 1010
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([482])
```

```
🔹 Batch 3
Batch DataBatch(x=[474, 5], edge_index=[2, 974], y=[32], batch=[474], ptr=[33])
Number of nodes: 474
Number of edges: 974
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([474])

🔹 Batch 4
Batch DataBatch(x=[517, 5], edge_index=[2, 1098], y=[32], batch=[517], ptr=[33])
Number of nodes: 517
Number of edges: 1098
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([517])

🔹 Batch 5
Batch DataBatch(x=[458, 5], edge_index=[2, 964], y=[32], batch=[458], ptr=[33])
Number of nodes: 458
Number of edges: 964
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([458])
```

```
for i, batch in enumerate(test_loader):
    print(f"\n🔹 Batch {i+1}")
    print(f"Batch {batch}")
    print("Number of nodes:", batch.x.size(0))
    print("Number of edges:", batch.edge_index.size(1))
    print("Target shape:", batch.y.shape)
    print("Graphs in batch:", batch.num_graphs)
    print("Batch vector shape:", batch.batch.shape)

    if i == 4:   # Stop after 5 batches
        break
```

```
🔹 Batch 1
Batch DataBatch(x=[655, 5], edge_index=[2, 1408], y=[32], batch=[655], ptr=[33])
Number of nodes: 655
Number of edges: 1408
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([655])

🔹 Batch 2
Batch DataBatch(x=[821, 5], edge_index=[2, 1824], y=[32], batch=[821], ptr=[33])
Number of nodes: 821
Number of edges: 1824
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([821])

🔹 Batch 3
Batch DataBatch(x=[722, 5], edge_index=[2, 1592], y=[32], batch=[722], ptr=[33])
Number of nodes: 722
Number of edges: 1592
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([722])

🔹 Batch 4
Batch DataBatch(x=[587, 5], edge_index=[2, 1246], y=[32], batch=[587], ptr=[33])
Number of nodes: 587
Number of edges: 1246
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([587])

🔹 Batch 5
Batch DataBatch(x=[649, 5], edge_index=[2, 1390], y=[32], batch=[649], ptr=[33])
Number of nodes: 649
Number of edges: 1390
Target shape: torch.Size([32])
Graphs in batch: 32
Batch vector shape: torch.Size([649])
```

```
import torch
device = "cuda" if torch.cuda.is_available() else "cpu"
device
```

```
'cuda'
```

```
from torch import nn
from torch.nn import Linear
import torch.nn.functional as F
from torch_geometric.nn import GCNConv
from torch_geometric.nn import global_mean_pool as gmp, global_max_pool as gap
```

Model_0:(Took the same model used for ESOL dataset)

```
class GCNModelV0(torch.nn.Module):
  def __init__(self, input_unit:int, hidden_unit:int, output_unit:int):
    super().__init__()
    self.conv1 = GCNConv(input_unit, hidden_unit)
    self.conv2 = GCNConv(hidden_unit, hidden_unit)
    self.conv3 = GCNConv(hidden_unit, hidden_unit)
    self.conv4 = GCNConv(hidden_unit, hidden_unit)
    self.linear = Linear(hidden_unit*2, 1)


  def forward(self, x, edge_index, batch):
    x = self.conv1(x, edge_index)
    x = F.tanh(x)
    x = self.conv2(x, edge_index)
    x = F.tanh(x)
    x = self.conv3(x, edge_index)
    x = F.tanh(x)
    x = self.conv4(x, edge_index)
    x = F.tanh(x)
```

```
    x = F.tanh(x)

    x = torch.cat([gmp(x, batch),
                           gap(x, batch)], dim=1)



    #x = F.dropout(x, p=0.5, training=self.training)

    x = self.linear(x)
    return x
```

```
torch.manual_seed(42)
model_0 = GCNModelV0(input_unit= 5, hidden_unit= 64, output_unit= 1)
print (model_0)
```

```
     GCNModelV0(
       (conv1): GCNConv(5, 64)
       (conv2): GCNConv(64, 64)
       (conv3): GCNConv(64, 64)
       (conv4): GCNConv(64, 64)
       (linear): Linear(in_features=128, out_features=1, bias=True)
     )
```

```
loss_fn = nn.MSELoss()
optimizer = torch.optim.Adam(params = model_0.parameters(), lr = 0.0007)
```

```
from timeit import default_timer as timer
def print_train_time(start: float, end: float):
    train_time =  end - start
    print(f"Total time taken for training: {train_time:.6f}seconds")
    return train_time
```

```
def train_loop(model: torch.nn.Module,
               dataloader: torch.utils.data.DataLoader,
               loss_fn: torch.nn.Module,
               optimizer: torch.optim.Optimizer):
    train_loss = 0

    for batch in train_loader:
        model.train()

        ##Forwars pass
        y_pred = model(batch.x.float(), batch.edge_index, batch.batch)

        ## Calculate the loss
        loss = loss_fn(y_pred, batch.y.float())
        train_loss += loss


        ## Optimizer zero grad
        optimizer.zero_grad()

        ## Loss backward
        loss.backward()

        ## optimer step
        optimizer.step()

    train_loss /= len(train_loader)

    print(f" Train loss: {train_loss}")


def test_loop(model: torch.nn.Module,
              dataloader: torch.utils.data.DataLoader,
              loss_fn: torch.nn.Module,
              optimizer: torch.optim.Optimizer):
    test_loss = 0
```

```
    for batch in test_loader:

        test_pred = model(batch.x.float(), batch.edge_index, batch.batch)

        ## Calculate loss
        loss = loss_fn(test_pred, batch.y.float())
        test_loss += loss


    test_loss /= len(test_loader)

    print(f"Test_loss: {test_loss}")



from tqdm.auto import tqdm ## FOR KEEPING THE progress bar
torch.manual_seed(42)
start_time = timer()
epochs = 20000

for epoch in tqdm(range(epochs)):
  if epoch % 1000 == 0:
    print(f"Epoch: {epoch}------------")
    train_loop(model_0, train_loader, loss_fn, optimizer)


    model_0.eval()
    with torch.inference_mode():
        test_loop(model_0, test_loader, loss_fn, optimizer)

end_time = timer()
total_time = end_time - start_time
print(f"The total time taken for the model to run is: {total_time}")
```

```
100%|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
20000/20000 [02:26<00:00, 136.24it/s]
```

```
Epoch: 0------------
/usr/local/lib/python3.11/dist-packages/torch/nn/modules/loss.py:610: UserWarning: Using a target size (torch.Size
  return F.mse_loss(input, target, reduction=self.reduction)
/usr/local/lib/python3.11/dist-packages/torch/nn/modules/loss.py:610: UserWarning: Using a target size (torch.Size
  return F.mse_loss(input, target, reduction=self.reduction)
/usr/local/lib/python3.11/dist-packages/torch/nn/modules/loss.py:610: UserWarning: Using a target size (torch.Size
  return F.mse_loss(input, target, reduction=self.reduction)
 Train loss: 12969.7734375
Test_loss: 13503.5625
Epoch: 1000------------
 Train loss: 9317.908203125
Test_loss: 10214.0185546875
Epoch: 2000------------
 Train loss: 8885.3359375
Test_loss: 9558.7294921875
Epoch: 3000------------
 Train loss: 8877.80859375
Test_loss: 9446.337890625
Epoch: 4000------------
 Train loss: 8871.73046875
Test_loss: 9437.025390625
Epoch: 5000------------
 Train loss: 8875.146484375
Test_loss: 9488.4658203125
Epoch: 6000------------
 Train loss: 8868.7548828125
Test_loss: 9497.04296875
Epoch: 7000------------
 Train loss: 8868.296875
Test_loss: 9474.45703125
Epoch: 8000------------
 Train loss: 8875.0322265625
Test_loss: 9526.263671875
Epoch: 9000------------
 Train loss: 8872.7353515625
Test_loss: 9507.78125
Epoch: 10000------------
 Train loss: 8880.234375
Test loss: 9493.1826171875
```

```
Epoch: 11000------------
 Train loss: 8870.13671875
Test_loss: 9522.7607421875
Epoch: 12000------------
 Train loss: 8873.138671875
Test_loss: 9506.908203125
Epoch: 13000------------
 Train loss: 8877.064453125
Test_loss: 9466.5419921875
Epoch: 14000------------
 Train loss: 8878.115234375
Test_loss: 9558.0517578125
Epoch: 15000------------
 Train loss: 8867.8642578125
Test_loss: 9475.787109375
Epoch: 16000------------
 Train loss: 8878.3388671875
Test_loss: 9645.3076171875
Epoch: 17000------------
 Train loss: 8884.287109375
Test_loss: 9464.7958984375
Epoch: 18000------------
 Train loss: 8873.666015625
Test_loss: 9456.5029296875
Epoch: 19000------------
 Train loss: 8877.9716796875
Test_loss: 9523.53125
The total time taken for the model to run is: 146.51956883399998
```

My God!!!! Such a poor model.... Definitely needto improve it

Start coding or generate with AI.